Growth, collisions and lamplighter structure of generic triangle and orthoscheme reflection groups

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Abstract

This paper consolidates a group of preprints on the reflection groups of generic Euclidean orthoschemes and their growth. The unconditional core is: the right-angled Coxeter envelope W_n has rational growth $(1+t)^{\lfloor n/2\rfloor+1}/K_{n+1}(t)$ with rate $r_n = 1+2\cos(2\pi/(n+3))$; a dichotomy identifies the collision depth of a leg tuple with half the shortest kernel length; for $n \geq 3$ every length-six kernel element lies in a rank-two standard parabolic and occurs only at dihedral angle $\pi/3$; over $\mathbb{Q}^n_{>0}$ the rank-two kernel condition is decided by the positional statistics of the leg tuple, the Diophantine input being three rank-zero elliptic curves, and the corresponding statement over $\mathbb{R}$ is false. In the plane the kernel of the triangle representation is nontrivial and the growth series of the generic triangle group departs from the envelope at depth 11. The growth rate of the planar right-triangle group is $\beta_2 = 1/\sqrt{q^*} = 1.4916177871\ldots$, where q^* is the smallest positive zero of a Hahn–Exton q-cosine; $\beta_2 < 3/2$ is unconditional. The travel resolvent is transcendental over $\mathbb{Q}(x)$ unconditionally.

Section 3 contains the one statement here that is not in the source preprints: for every $n \geq 2$, any embedding of W_n into $\mathrm{Isom}(\mathbb{R}^n)$ has injective linear part, so the ambient embedding problem is equivalent to the question of whether W_n embeds in the *compact* group $O(n)$. At $n = 2$ this recovers the planar obstruction from solvability of $O(2)$ alone, without amenability of $\mathrm{Isom}(\mathbb{R}^2)$, and it isolates non-commutativity of $O(n)$ as the exact point at which the planar argument fails for $n \geq 3$.

Section 1 states, for each result, what it is conditional on. Three hypotheses are open and are named wherever they are used: the strand-walk transfer model (M), the junction pairing $(R{-}J)$, and the invariance of the travel block (T). Transcendence of the full growth series U and of its relaxed companion V is conditional on these and is not claimed here. Irrationality of β_2 is open.

Section 4 settles one of the open problems of this circle in the negative: the conjecture that the orthoscheme representation is faithful at every Class C rational leg tuple is false, with explicit counterexamples in dimension three, the shortest kernel element found having length 76.

1 What is new, what is known, and what is conditional

1.1 Prior art

The following are not ours and are used as such. Isola and Piergallini proved that the generic triangle reflection group has free metabelian rotation subgroup of rank two and is not finitely presented. Droms, Lewin and Servatius, and Myasnikov, Roman'kov, Ushakov and Vershik, gave the length formula $\|\phi\|_1 + 2\,\mathrm{st}(\phi)$ for free solvable quotients F/N'; Sale gave the metric behaviour of the Magnus embedding. Steinberg's formula for Poincaré series of Coxeter groups is standard, as are the independence polynomials of paths and the location of their roots. Adamczewski, Dreyfus and Hardouin proved hypertranscendence of the solutions of the relevant q-difference equation for all q not a root of unity, which is stronger than the SL_2 statement recorded in §8; our increment there is only the removal of a transcendence hypothesis on q in that one special case.

1.2 Claim labels

Every numbered statement below carries exactly one of PROVED (complete human proof), VERIFIED (formalized in Lean 4, axioms audited), HEURISTIC (computational support only), or CONJECTURE. The effective strength of a result is the weakest label in its dependency chain.

1.3 The three open hypotheses

Hypothesis 1.1 (transfer model, (M))**.** The strand-walk model computes the relaxed word length, that is, the crossing optimisation of the site-cost chain is the relaxed length.

Hypothesis 1.2 (junction pairing, $(R{-}J)$)**.** The junction pairing $\Pi_y(q_m)$ is nonzero at infinitely many travel poles q_m.

Hypothesis 1.3 (travel-block invariance, (T))**.** The denominator $1 - \Sigma_1$ is the identical, cycle-marker-free object in the generating functions of U and of V, so U and V share the travel poles.

Hypothesis 1.3 is HEURISTIC. Its strand-walk argument rests on the localisation of cycles, checked exhaustively on the 3869 pure-travel elements of true length at most 16, and on the metric identity $\ell_T = \ell_R + 2c$, whose upper bound is PROVED and whose lower bound is open. Hypothesis 1.2 is CONJECTURE. Hypothesis 1.1 is HEURISTIC and everything in the block-assembly sections is conditional on it.

Remark 1.4 (what is not claimed). Three claims that appeared in a draft consolidation are withdrawn here because they are not supported by the sources.

(i) Irrationality of β_2 is *open*. It is not proved, and no effective irrationality measure for β_2 is available. What is proved is $\beta_2 \neq 3/2$, by the squeeze of §6. The reformulation known is that β_2 is irrational if and only if a Hahn–Exton function omits the value 0 at the relevant rational points; that is a restatement, not a proof, and it supplies no exponent.

(ii) Hypothesis 1.3 is *not* a theorem.

(iii) The lower bound of the metric identity $\ell_T = \ell_R + 2c$ is *open*; only the upper bound is PROVED. Statements that consume the identity inherit that standing.

2 The envelope and the dichotomy

Let $O(a_1, \ldots, a_n) \subset \mathbb{R}^n$ be the right-corner orthoscheme with vertices $V_0 = 0$ and $V_k = V_{k-1} + a_k e_k$, and let $\widehat{R}_0, \ldots, \widehat{R}_n$ be the reflections in its facets, generating $W(a) \leq \mathrm{Isom}(\mathbb{R}^n)$. Facets i, j with $|i - j| \geq 2$ are perpendicular for every leg tuple, so with Γ_n the graph on $\{0, \ldots, n\}$ having an edge whenever $|i - j| \geq 2$,

$$W_n = \big\langle R_0, \ldots, R_n \;\big|\; R_i^2 = 1,\ (R_i R_j)^2 = 1\ (|i - j| \geq 2) \big\rangle \tag{1}$$

is the right-angled Coxeter group on Γ_n, and $\rho_a \colon W_n \twoheadrightarrow W(a)$.

Theorem 2.1 (Envelope; PROVED). *$W_n(t) = (1+t)^{\lfloor n/2\rfloor+1}/K_{n+1}(t)$, with $K_0 = K_1 = 1$, $K_{2j} = K_{2j-1} - tK_{2j-2}$ and $K_{2j+1} = (1+t)K_{2j} - tK_{2j-1}$. The growth rate is $r_n = 1 + 2\cos(2\pi/(n+3))$.*

Theorem 2.2 (Dichotomy; PROVED). *For a marked quotient $Q = G/N$ with $K(N) = \min_{1\neq g\in N} \ell_G(g)$ and $h = \lceil K/2\rceil$, the sphere sizes satisfy $s_d(Q) = s_d(G)$ for $d < h$ and $s_h(Q) < s_h(G)$. Hence $\mathrm{cd}_n(a) = K(\ker\rho_a)/2$.*

Theorem 2.3 (Length six; PROVED). *For $n \geq 3$ and $a \in \mathbb{R}^n_{>0}$, if $w \in \ker\rho_a$ has $\ell_{W_n}(w) = 6$ then $w = (R_iR_{i+1})^{\pm 3}$ for some i with $\theta_{i,i+1} = \pi/3$.*

Theorem 2.4 (Rank-two classification; PROVED). *For $a \in \mathbb{Q}^n_{>0}$, $\ker\rho_a$ meets a rank-two standard parabolic nontrivially if and only if $e(a)+t(a) \geq 1$, where e counts endpoint equal pairs and t counts three consecutive equal legs. The Diophantine input is the triviality of the rational points on three quartics with Jacobians 24a1, 72a2 and $y^2 = x^3 - x$, all of rank zero. Over $\mathbb{R}$ the statement is false: $a = (\sqrt{3}, 1, 5)$ has a rank-two kernel element with no equal legs.*

3 The ambient embedding reduces to a compact group

This section is the one place where the present paper goes beyond the sources. Write $L\colon \mathrm{Isom}(\mathbb{R}^n) \to O(n)$ for the linear part, with kernel the translation group $\mathbb{R}^n$.

Lemma 3.1 (PROVED). *For every $n \geq 2$, W_n has no nontrivial normal abelian subgroup.*

Proof. The Coxeter diagram of the system (1) has an edge exactly where $m_{ij} \geq 3$, that is, exactly where $\{i, j\}$ is a non-edge of Γ_n, that is, exactly where $|i-j| = 1$. So the diagram is the path on $n+1$ nodes with every edge labelled ∞. The path is connected, so (W_n, S) is irreducible. Every label is ∞, so W_n is infinite and is not of affine type, affine Coxeter systems having all labels finite.

Let $A \trianglelefteq W_n$ be abelian; A is amenable, so it lies in the amenable radical of W_n. We show that radical is trivial, in two steps.

First it is finite. An irreducible, infinite, non-affine Coxeter group contains a rank-one isometry for its action on the Davis complex (Caprace–Fujiwara), and a group acting properly cocompactly on a CAT(0) space with a rank-one isometry, and which is not virtually cyclic, is acylindrically hyperbolic. W_n is not virtually cyclic: for $n \geq 3$ it contains $\langle R_0, R_2\rangle \times \langle R_1\rangle$, and for $n = 2$ it is $(\mathbb{Z}/2 \times \mathbb{Z}/2) * \mathbb{Z}/2$, which contains a free group of rank two. For an acylindrically hyperbolic group the amenable radical coincides with the finite radical, the unique maximal finite normal subgroup (Osin). Note that acylindrical hyperbolicity gives finiteness here, not triviality; the second step is needed.

Second it is trivial. The finite radical of a Coxeter group is the direct product of its finite irreducible components, and W_n is irreducible and infinite, so it has no finite component and its finite radical is trivial. Hence $A = 1$. $\square$

Proposition 3.2 (Reduction to $O(n)$; PROVED). *Let $n \geq 2$ and let $\varphi\colon W_n \to \mathrm{Isom}(\mathbb{R}^n)$ be an injective homomorphism. Then $L \circ \varphi$ is injective. Consequently*

$$W_n \textit{ embeds in } \mathrm{Isom}(\mathbb{R}^n) \iff W_n \textit{ embeds in } O(n).$$

Proof. $\ker(L\circ\varphi) = \varphi^{-1}\big(\varphi(W_n) \cap \mathbb{R}^n\big)$ is normal in W_n and is isomorphic to a subgroup of the abelian group $\mathbb{R}^n$, hence abelian. By Lemma 3.1 it is trivial. The equivalence follows, one direction being $O(n) \leq \mathrm{Isom}(\mathbb{R}^n)$. $\square$

Corollary 3.3 (The plane, without amenability; PROVED). *W_2 does not embed in* $\mathrm{Isom}(\mathbb{R}^2)$.

Proof. $O(2) = SO(2) \rtimes \mathbb{Z}/2$ is metabelian, hence solvable, hence contains no free subgroup of rank two. $W_2 = (\mathbb{Z}/2 \times \mathbb{Z}/2) * \mathbb{Z}/2$ is a free product of finite groups of orders 4 and 2 and is not infinite dihedral, so it contains a free subgroup of rank two. So W_2 does not embed in $O(2)$, and Proposition 3.2 transfers this to $\mathrm{Isom}(\mathbb{R}^2)$. □

Remark 3.4 (where the planar argument really fails). The published planar obstruction runs through amenability of $\mathrm{Isom}(\mathbb{R}^2)$. That route is unavailable for $n \geq 3$ because $SO(3)$ contains a free subgroup of rank two, so $\mathrm{Isom}(\mathbb{R}^n)$ is not amenable as an abstract group. Proposition 3.2 relocates the obstruction: the translation part is never the issue in any dimension, and the planar proof uses exactly one property of the ambient group, namely that $O(2)$ is solvable. For $n \geq 3$ the target $O(n)$ is a compact, non-solvable Lie group, and that single change is the whole of the difficulty.

Remark 3.5 (what the reduction buys). Replacing $\mathrm{Isom}(\mathbb{R}^n)$ by $O(n)$ replaces a non-compact target by a compact one, and compactness is usable. If $W_n \leq O(n)$, its closure H is a compact Lie subgroup of $O(n)$, W_n is dense in H, and H is infinite since W_n is. Hence H° is a connected compact Lie group of dimension at most $n(n-1)/2$, the subgroup $W_n \cap H^\circ$ has finite index in W_n and is dense in H°. In particular an embedding would give W_n a faithful orthogonal representation of degree n. This is a much more rigid demand than a faithful action by isometries of $\mathbb{R}^n$, and it is the form in which we propose Problem 9.4 be attacked.

Remark 3.6 (the point group, measured). Proposition 3.2 says the linear part carries everything, so the object to measure is the point group $\pi \circ \rho_a \colon W_n \to O(n)$, not ρ_a. If the point group is injective then so is ρ_a, so a faithful point group would settle Problems 9.2, 9.4 and 9.5 together; the converse fails, and a deviation of the point group is not a deviation of ρ_a. The point group has not previously been measured on its own. Running it against the envelope at two independent primes $2^{61} - 1$ and $2^{62} - 57$, at Class C leg tuples, gives agreement to the following depths:

n	2	3	4	5	6
legs	$(1, 2)$	$(1, 2, 3)$	$(1, 2, 3, 5)$	$(3, 1, 4, 2, 5)$	$(1, 2, 3, 5, 7, 4)$
agrees to depth	**2**	22	17	15	13

and a second tuple at $n = 3$, namely $(2, 5, 3)$, and at $n = 4$, namely $(3, 1, 4, 2)$, give the same depths as the first. The $n = 2$ column is a negative control and it fires: there the point group lies in $O(2)$, which is solvable, while W_2 contains a free group of rank two, so a deviation is forced, and it occurs at depth 3, the earliest depth not already fixed by the $(R_0R_2)^2 = 1$ relation. For $n \geq 3$ no deviation is found. This is HEURISTIC and decides nothing: it is consistent both with faithfulness and with a horizon beyond the computed range. What it does establish, at each tuple listed, is that the point group is injective on the ball of the stated radius, hence so is ρ_a; at $n = 3$ that improves the previously recorded depth for ρ_a from 19 to 22, and does so for the stronger object.

4 Class C faithfulness is false

Remark 3.6 found Class C tuples at which the point group is *not* injective. That is not by itself a statement about ρ_a, the point group being a quotient. The following elementary observation converts it into one, and needs neither Lemma 3.1 nor any CAT(0) input.

Proposition 4.1 (Translations commute; PROVED)**.** *Let $a \in \mathbb{R}^n_{>0}$ and let $u \in W_n$ satisfy $\pi(\rho_a(u)) = I$, that is, $\rho_a(u)$ has trivial linear part. Then for every $g \in W_n$,*

$$[\, u,\ gug^{-1}\,] \;\in\; \ker \rho_a.$$

Proof. An isometry of $\mathbb{R}^n$ with trivial linear part is a translation, so $\rho_a(u)$ is a translation. For any g, $\rho_a(gug^{-1}) = \rho_a(g)\rho_a(u)\rho_a(g)^{-1}$ is a conjugate of a translation by an isometry, hence again a translation. Two translations of $\mathbb{R}^n$ commute, so the commutator of their preimages is killed. $\square$

Theorem 4.2 (Conjecture C is false; PROVED)**.** *There are Class C tuples $a \in \mathbb{Q}^3_{>0}$, that is with pairwise distinct coordinates, at which ρ_a is not injective. Explicitly this holds at*

$$(1,3,2), \quad (2,3,9), \quad (2,6,9), \quad (3,1,5), \quad (3,2,6), \quad (4,9,7), \quad (6,11,7)$$

and at their reversals. At $a = (3,1,5)$ an explicit nontrivial element of $\ker \rho_a$ has length 76, obtained by reducing an explicit word of length 92.

Proof. Take $a = (3,1,5)$; the other tuples are identical in form. Breadth-first search in W_3, using the right-angled normal form and exact rational arithmetic for the linear reflections, produces two elements $w_1 \neq w_2$ of W_3 of length 11 with the same linear part; the sphere count of the point group at depth 11 falls short of the envelope, at each of two independent primes. Put $u = w_1 w_2^{-1}$, of length 22. Then $\pi(\rho_a(u)) = I$ by construction, and $u \neq 1$ in W_3, verified in the Tits geometric representation of the Coxeter system, which is faithful for every Coxeter system and has integer matrices here, so the verification is exact. By Proposition 4.1, $c = [u,\, R_1 u R_1]$ lies in $\ker \rho_a$, and $c \neq 1$ in W_3, again by the Tits representation; reducing c gives a geodesic of length 76. Finally the coordinates of a are pairwise distinct, so a is Class C. $\square$

Remark 4.3 (why this was not seen)*.* The counterexamples are invisible to a direct search. By the dichotomy the deviation of ρ_a sits at depth $\lceil K/2 \rceil$, and here $K \leq 76$, so the first depth at which the orbit growth of $W(a)$ departs from the envelope is somewhere below 38 but far above any reachable breadth-first search: the envelope sphere at depth 38 is $9 \cdot 2^{36}$, already above $6 \cdot 10^{11}$. A direct computation on ρ_a at these tuples sees agreement with the envelope for as long as it can run, and indeed does. What makes the counterexamples accessible is that the *linear part* fails much earlier, at depth 11 to 13, where the search is trivial. That is the practical content of Proposition 3.2: the obstruction lives in $O(n)$, and it can be seen there.

Corollary 4.4 (a necessary condition; PROVED)**.** *If ρ_a is injective then the point group $\pi \circ \rho_a$ is injective. Equivalently, a point-group collision at a certifies that ρ_a is not injective.*

Proof. $\ker(\pi \circ \rho_a)$ is normal in W_n, and if ρ_a is injective it is isomorphic to its image, a subgroup of the translation group, hence abelian. By Lemma 3.1 it is trivial. $\square$

Remark 4.5 (what remains open)*.* Theorem 4.2 settles the arithmetic form of Problem 9.2 negatively. It does *not* settle the generic form: the tuples above are single points, and $N_{\text{gen}} = \bigcap_a \ker \rho_a$ may still be trivial, since a kernel element at one tuple need not be a kernel element at another. It also leaves Problem 9.4 open, an abstract embedding not being required to come from an orthoscheme.

Remark 4.6 (what the reduction does not buy)*.* It does not decide Problem 9.4 for any $n \geq 3$, and it does not touch Problem 9.2: an abstract embedding of W_n into $O(n)$ need not be by reflections, and the faithfulness of the particular map ρ_a is a separate question. The two standard invariants remain undecisive. Finite subgroups do not separate: every finite subgroup of W_n is conjugate into an elementary abelian 2-group on a clique of Γ_n, of rank at most $\lceil (n+1)/2 \rceil \leq n$, while $O(n)$ contains $(\mathbb{Z}/2)^n$. Free subgroups do not separate either, both groups containing free subgroups of rank two for $n \geq 3$.

5 The planar case

Let $F = C_2 * C_2 * C_2$ and $\rho_\tau \colon F \to G_\tau$ the generic triangle group.

Theorem 5.1 (Census; PROVED, computational input VERIFIED)**.** *There are* 33 *pairs of reduced words of length* 11*, distinct in* F*, with* $\rho_\tau(w) = \rho_\tau(w')$ *for every* τ*. Outside a proper Zariski-closed subset of shape space these are the only coincidences in the ball of radius* 11*, and there are* 132 *further pairs at radius* 12*. The generic sphere sizes are*

$$1, 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536, 3039, 6012, 11925, 23570.$$

The 33 identities are polynomial identities in $\mathbb{Z}[p, q]$; the lower bounds come from counts in a finite field chosen to contain the needed roots of unity and $\sqrt{-1}$, and match the upper bounds from the identities, so the counts are exact.

Theorem 5.2 (Right-triangle structure; PROVED)**.** *For* $W = \langle R_0, R_1, R_2 \mid R_i^2 = (R_0R_1)^2 = 1\rangle$ *the rotation subgroup is of index* 4 *and isomorphic to* $\mathbb{Z}\wr\mathbb{Z}$*, with translation lattice* $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}]$ *obtained from Crowell's exact sequence and the Bass–Serre tree of the free product. The universal sequence is shape-independent through depth* 32*, and the shape* $(1, 2)$ *deviates at depth* 33*. If* $c_T = N(a - ib) > 2d$ *then no deviation occurs at depth* d*.*

Remark 5.3 (status of the metric)*.* Word length equals relaxed length plus twice the connectivity defect, $\ell_T = \ell_R + 2c$. The upper bound is PROVED. The lower bound is open and is stated as a conjecture; the closed form c_{pred} computed by the explicit transducer agrees with the true defect with 0 exceptions on the 275,823 elements of the ball of radius 24, held out at depths $19, 21, 23, 24$, which is HEURISTIC support and not a proof. The junction site cost is $\max(|d_L - 1|, |d_R|)$, which depends on the sign of d_L and not on its magnitude alone; the form $\max(|d_L| - 1, |d_R|)$ is false on every cell with $d_L \le -2$ and $|d_R| \le |d_L|$. This is a term inside the site-cost chain of the strand model, not an additive correction to the length formula.

6 The growth rate β_2

With $q = x^2$ and Σ_1 the odd k-recursion sum, $1 - \Sigma_1(q)$ is a Hahn–Exton q-cosine, and $q^* = 0.449453630558948046\ldots$ is its smallest positive zero.

Theorem 6.1 (PROVED)**.** *The growth rate of the planar right-triangle group is* $\beta_2 = 1/\sqrt{q^*} = 1.4916177871\ldots$*, and* $\beta_2 < 3/2$*.*

The proof squeezes the true count between the pure-travel and relaxed counts, which share the exponential rate, and locates q^* by a contour argument with exact rational winding number. Irrationality of β_2 is open (Remark 1.4).

7 Transcendence: the unconditional part

Theorem 7.1 (PROVED)**.** Σ_1 *and* S_1 *are transcendental over* $\mathbb{Q}(q)$ *and have* $|q| = 1$ *as a natural boundary.*

Theorem 7.2 (Travel resolvent; PROVED)**.** $\Sigma_0/(1 - \Sigma_1)$ *is transcendental over* $\mathbb{Q}(x)$*, unconditionally.*

Transcendence of V is conditional on (M) and $(R{-}J)$; transcendence of U is conditional on (M), $(R{-}J)$ and (T). Neither is claimed here.

8 The q-cosine increment

Theorem 8.1 (PROVED)**.** *For every algebraic $q \in (0,1)$ the difference Galois group of $G(q,z) = {}_1\phi_1(0;q;q^2,z)$ over $\mathbb{C}(z)$ contains SL_2, so G is hypertranscendental. The connection constant is $C_1 = 1/(q;q)_\infty$, whence $\prod_{k\geq 1} z_k(q)q^{2(k-1)} = (q;q^2)_\infty$.*

The general hypertranscendence statement, for all q not a root of unity and over $\bigcup_j \mathbb{C}(z^{1/j})$, is due to Adamczewski, Dreyfus and Hardouin and is stronger. The increment is the removal of a transcendence hypothesis on q in the SL_2 statement.

9 The five directions

Problem 9.1 (Prove $(R$–$J)$)**.** Show $\Pi_y(q_m) \neq 0$ for infinitely many travel poles. This would make V, and with (T) also U, transcendental over $\mathbb{Q}(x)$ unconditionally, modulo (M). What exists is an upper bound $|P_{12}|w^3 < 2$; what is needed is a lower bound $|P_{12}(q_m)| \geq c\, w_m^{-3}$.

Problem 9.2 (The alternative in dimension $n \geq 3$)**.** Is $N_{\text{gen}} = \bigcap_a \ker \rho_a$ trivial, equivalently is ρ_a injective for generic a? A single faithful tuple answers this affirmatively. The arithmetic form is FALSE (Theorem 4.2), which does not decide the generic form.

Remark 9.3 (Class C faithfulness)*.* The conjecture that ρ_a is injective for every $a \in \mathbb{Q}^n_{>0}$ with $e(a) = t(a) = 0$ and $n \geq 3$ is FALSE, by Theorem 4.2. Only the generic form (a) survives.

Problem 9.4 (Ambient embedding)**.** For $n \geq 3$, does W_n embed in $O(n)$? By Proposition 3.2 this is equivalent to the ambient embedding question for $\operatorname{Isom}(\mathbb{R}^n)$, and by Remark 3.5 it asks whether W_n has a faithful orthogonal representation of degree n.

Problem 9.5 (Finite presentation)**.** For $n \geq 3$, is W_n/N_{gen} finitely presented? A negative answer refutes Problem 9.2(a). At $n = 2$ the answer is no.

Problem 9.6 (Transcendence of the zeros)**.** Are the zeros $z_k(q)$ of the Hahn–Exton q-cosine transcendental at algebraic $q \in (0,1)$, and is β_2 transcendental? This is the q-analogue of Siegel's theorem on the zeros of Bessel functions. Even irrationality of β_2 is open.